IIT ROORKEE

FREE ONLINE EDUCATION swayam शिक्षित भारत, उन्नत भारत

NPTEL ONLINE CERTIFICATION COURSE

Charging Infrastructure

Lecture-39

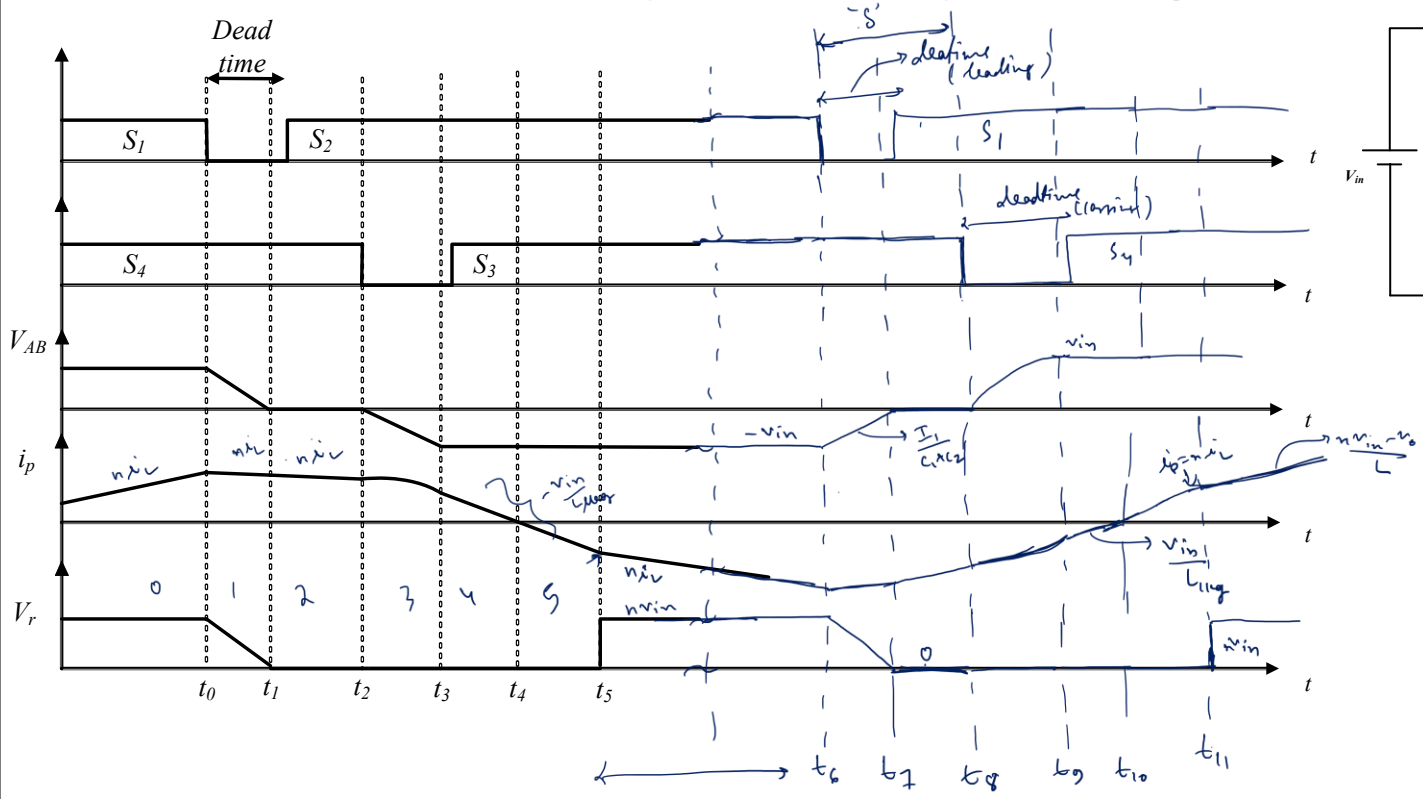
Small Signal Modelling of PSFB-I

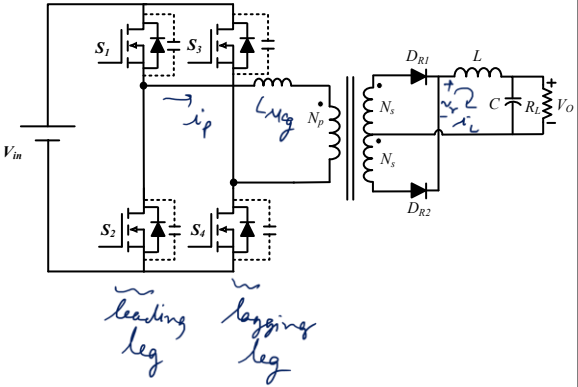
Dr. Apurv Kumar Yadav

Department of Electrical Engineering

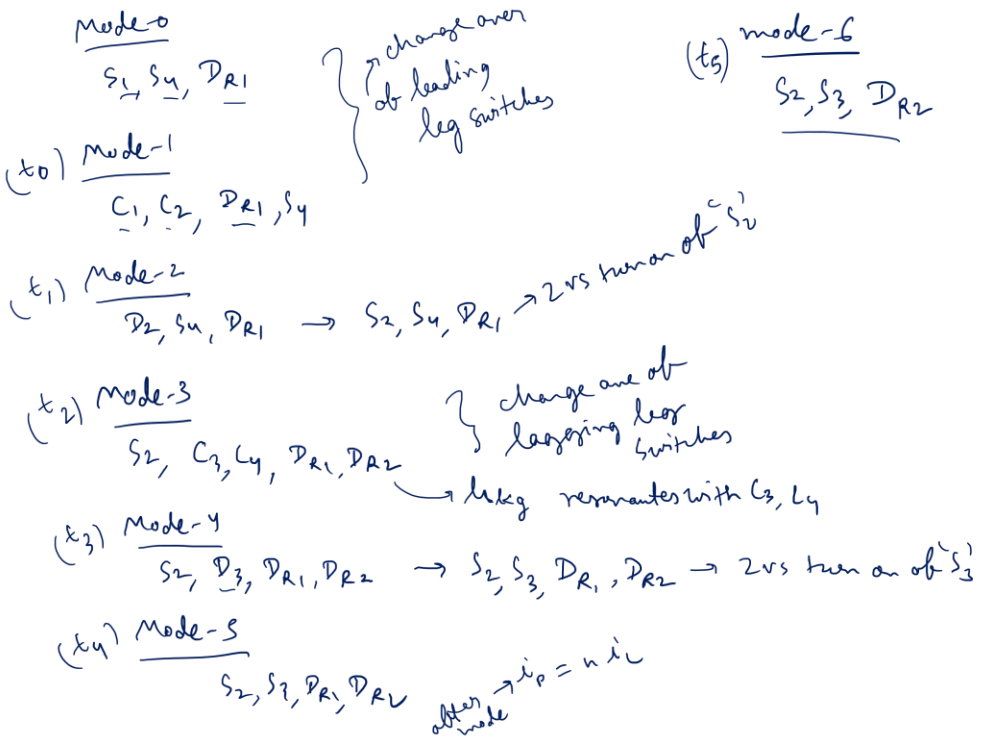


Recap: PSFB Operating Modes





Recap



Mode-3 to Mode-5

$t_2 - t_5$ are the duty loss duration

required → $t_{deadtime(leading)} > t_{01}$

$> \frac{2CV_{in}}{I_1}$ $\quad (C = C_1 = C_2)$

required → $t_{deadtime(lagging)} > t_{23}$

$> \frac{1}{\omega_1} \sin^{-1}\left(\frac{V_{in}}{I_2}\sqrt{\frac{2C}{L_{eq}}}\right)$

Recap

necessary condition for ZVS of leading leg (turn-on)

$$\frac{1}{2}\left(L_{Mag}+\frac{L}{n^2}\right)I_1^2 > \frac{1}{2}C_1V_{in}^2+\frac{1}{2}C_2V_{in}^2 \longrightarrow$$

necessary condition for ZVS of lagging leg (turn-on)

$$\frac{1}{2}\left(L_{Mag}\right)\cdot i_p^2 > \frac{1}{2}C_3V_{in}^2+\frac{1}{2}C_4V_{in}^2$$

→ the lagging leg has problem in achieving ZVS during turn-on, when we have light load, small L_{Mag}, large C_3 & C_4, large input voltage

$$\text{Duty loss} = D_{loss} = \frac{t_5-t_2}{T_s/2} = \frac{t_{23}+t_{43}+t_{54}}{T_s/2}$$

t_{23} ≃ small values as compare to t_{43} & t_{54}

Recap

$$D_{loss} = \frac{4 L_{Mag} n I_o \cdot f_{sw}}{V_{in}} \Rightarrow V_o = n V_{in} (D - D_{loss})$$

$$\delta = \frac{t_2 - t_0}{T_{s/2}} \times 180^\circ$$

$$\frac{(1-D) T_s}{2} \approx \frac{\delta}{180^\circ} \cdot T_{s/2}$$

- to achieve $\rightarrow L_{Mag} \uparrow \rightarrow D_{loss} \uparrow$
  - $V_{in} \uparrow \rightarrow D_{loss} \downarrow$; $I_o \uparrow \rightarrow D_{loss} \uparrow$; $f_{sw} \uparrow \rightarrow D_{loss} \uparrow$
- If one wants to have ZVS of both the legs, they have compromise on D_{loss}

$$D_{eff} = D - D_{loss}$$

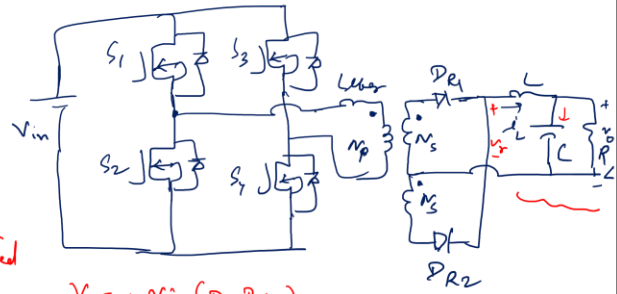
$$D_{loss} = \frac{4\, L_{leg}\, n I_o f_{sw}}{V_{in}} \simeq \frac{4\, L_{leg}\, n I_L f_{sw}}{V_{in}}$$

$$V_o = n V_{in} D_{eff}$$

$$\Rightarrow D_{eff} = \uparrow D - D_{loss} \uparrow$$

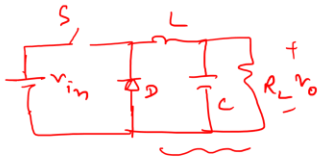
D_{eff} related to L_{leg}, f_{sw}, V_{in}, i_L

$\tilde{v}_{in}$ & $\tilde{i}_L \longrightarrow \tilde{d}_{eff}$



⇒ this is the buck derived topology

$V_o = n V_{in} (D - D_{loss})$





Control objective

① To regulate the output voltage as per the reference.

to regulate the o/p voltage, $\tilde{d} \rightarrow \tilde{i}_L \rightarrow \tilde{v}_o$

$\tilde{d}_i \leftarrow \tilde{d}_{loss}$

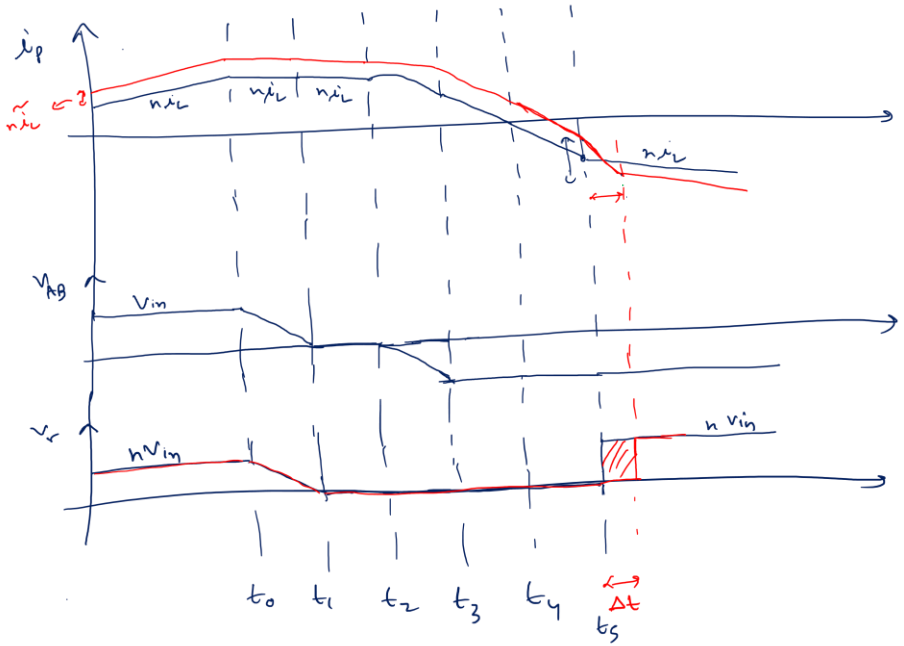
$\Rightarrow \tilde{d}_{eff} = \tilde{d} + \tilde{d}_i + \tilde{d}_v$

$\tilde{d}_i \rightarrow$ change in the duty ratio due to change in inductor current

$\tilde{d}_v \rightarrow$ change in the duty ratio due to change in input voltage

The dynamics in d_{eff} is the combined dynamics due to change in d, d_i, d_v

effect of $\tilde{d}_i$



$$t_{ns} = \frac{L_{mag}\, n I_L(t_5)}{v_{in}}$$

$$t_{ns,new} = \frac{L_{mag}\left(n I_L + n\tilde{i}_L\right)}{v_{in}}$$

$$t_{ns,old} = \frac{L_{mag}\left(n I_L - n\tilde{i}_L\right)}{v_{in}}$$

$$\Rightarrow \Delta t = \frac{L_{mag}\left(2n\tilde{i}_L\right)}{v_{in}}$$

$\tilde{i}_L \uparrow$, $\Delta t \downarrow$ (because the effect is opposite of case)

$\Delta t \uparrow$ corresponds to change in D_{eff} due to change in i_L

Source: J. A. Sabate, V. Vlatkovic, R. B. Ridley, F. C. Lee and B. H. Cho, "Design considerations for high-voltage high-power full-bridge zero-voltage-switched PWM converter," *Fifth Annual Proceedings on APEC*, Los Angeles, CA, USA, 1990, pp. 275-284, doi: 10.1109/APEC.1990.66420.

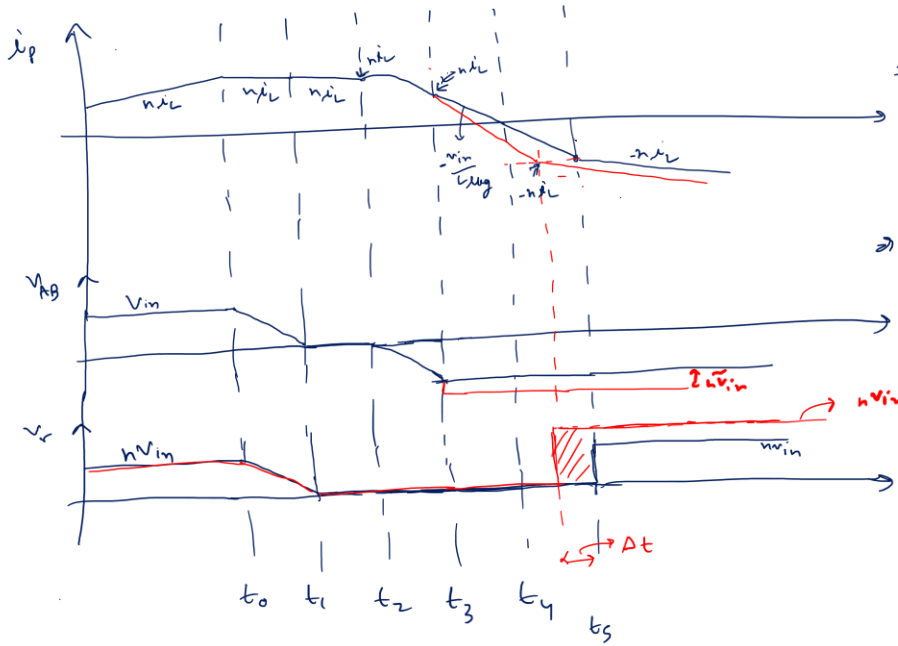
$$\Rightarrow \tilde{d}_i = -\frac{\Delta t}{T_s/2} = -\frac{L_{Mag}(2n\tilde{i}_v)}{v_{in} \times T_s} \times 2 = -\frac{4n L_{Mag} b_{sw} \tilde{i}_v}{v_{in}} \quad \left(T_s = \frac{1}{b_{sw}}\right)$$

$$\Rightarrow \boxed{\tilde{d}_i = -\frac{4n L_{Mag} b_{sw} \tilde{i}_v}{v_{in}}}$$

$\tilde{i}_v \uparrow, \Delta t \downarrow \Rightarrow$ the -ve sign is used

Source: J. A. Sabate, V. Vlatkovic, R. B. Ridley, F. C. Lee and B. H. Cho, "Design considerations for high-voltage high-power full-bridge zero-voltage-switched PWM converter," *Fifth Annual Proceedings on APEC*, Los Angeles, CA, USA, 1990, pp. 275-284, doi: 10.1109/APEC.1990.66420.

effect of $\tilde{v}_{in}$ (due to change in input voltage)



$(v = L\frac{di}{dt} \Rightarrow dt = \frac{L\,di}{v})$

if we assume t_{23} to be small (assumptions)

$$\Rightarrow \Delta t = \left[n i_L - \left(-n i_L + n \frac{V_o}{L} \frac{(1-D)T_s}{2} \right) \right] \left[\frac{L_{leg}}{V_{in}} - \frac{L_{leg}}{(V_{in}+\tilde{v}_{in})} \right]$$

($\frac{V_o}{L}\frac{(1-D)T_s}{2}$ is due to ripple in the output inductor current)

due to change in v_{in}

$$\Rightarrow \Delta t = \left[2nI_o - n\frac{V_o}{L}\frac{(1-D)T_s}{2} \right] \left[\frac{L_{leg}\tilde{v}_{in}}{V_{in}(V_{in}+\tilde{v}_{in})} \right]$$

Source: J. A. Sabate, V. Vlatkovic, R. B. Ridley, F. C. Lee and B. H. Cho, "Design considerations for high-voltage high-power full-bridge zero-voltage-switched PWM converter," *Fifth Annual Proceedings on APEC*, Los Angeles, CA, USA, 1990, pp. 275-284, doi: 10.1109/APEC.1990.66420.

$$\Delta t = n\left[2I_o - \frac{V_o}{L}\frac{(1-D)T_s}{2}\right]\left[\frac{L_{leg}\,\tilde{v}_{in}}{V_{in}(V_{in}+\tilde{v}_{in})}\right]$$

$\tilde{v}_{in}$ is small change $\Rightarrow$ $V_{in}+\tilde{v}_{in} = V_{in}$

$$\Rightarrow \Delta t = n\left[2I_o - \frac{V_o}{L}\frac{(1-D)T_s}{2}\right]\left[\frac{L_{leg}\,\tilde{v}_{in}}{V_{in}^2}\right] \longrightarrow (1)$$

$$\Rightarrow \tilde{d}_v = \frac{\Delta t}{T_s/2} = \frac{n\left[2I_o - \frac{V_o}{L}\frac{(1-D)T_s}{2}\right]\left[\frac{L_{leg}}{V_{in}^2}\right]\tilde{v}_{in}}{T_s/2}$$

$$\Rightarrow \tilde{d}_{eff} = \tilde{d} + \tilde{d}_i + \tilde{d}_v$$

Source: J. A. Sabate, V. Vlatkovic, R. B. Ridley, F. C. Lee and B. H. Cho, "Design considerations for high-voltage high-power full-bridge zero-voltage-switched PWM converter," *Fifth Annual Proceedings on APEC*, Los Angeles, CA, USA, 1990, pp. 275-284, doi: 10.1109/APEC.1990.66420.

Thank You